



Today I am learning to use probability to judge risk and uncertainty.

1. Understanding Risk Levels

Write each event in the correct risk category.

Low Risk Unlikely to happen	Medium Risk Could happen	High Risk Likely to happen	Very High Risk Almost certain

- | | |
|--|---|
| a) Getting sunburnt on a cloudy day. | f) It raining when you go outside during a storm. |
| b) Catching a cold in winter. | g) A new baby in your family. |
| c) Winning the lottery. | h) Getting tired after a long run. |
| d) Forgetting your homework. | i) A power cut during a severe storm. |
| e) Having an accident if you never cross the road. | j) Finding a coin on the ground. |

2. Judge the Risk

For each situation, estimate the probability as a fraction or percentage, then choose the best risk level.

Situation	Probability (Fraction or %)	Best Risk Level (circle one)			
		Low	Medium	High	Very High
a) The sun will rise tomorrow.	/ %	☐	☐	☐	☐
b) You will roll a 6 on a number cube.	/ %	☐	☐	☐	☐
c) You will get a head on a coin toss.	/ %	☐	☐	☐	☐
d) It will snow in summer where you live.	/ %	☐	☐	☐	☐
e) You will win first prize in a school raffle.	/ %	☐	☐	☐	☐

3. Compare Risks

For each pair of events, write <, > or = to compare the risk.

- | | | | |
|---|----------------------|---|----------------------|
| a) Getting a headache
or
Winning a gold medal in the Olympics | <input type="text"/> | d) Breaking a bone while playing safe
or
Getting hungry before dinner | <input type="text"/> |
| b) It raining today
or
Finding \$20 on the footpath | <input type="text"/> | e) Getting a flat tyre this week
or
Volcano eruption near your home | <input type="text"/> |
| c) Failing a spelling test
or
Earthquake tomorrow | <input type="text"/> | f) Losing a pencil today
or
Lightning strike in your yard | <input type="text"/> |

4. Risk in Real Life

Read each scenario and answer the questions.

- a) You are going on a hiking trip. The weather forecast says there is a 30% chance of rain.
- Is this a low, medium, high or very high risk of rain? _____
 - What will you do to prepare for this risk? _____
- b) A school is planning a sports day. The forecast shows an 80% chance of sunshine.
- Is this a low, medium, high or very high risk of rain? _____
 - How confident can the school be that sports day will go ahead? _____
- c) A bike rider wears a helmet. The chance of a serious head injury in a crash is reduced to 10%.
- Is the risk of serious head injury high or low with a helmet? _____
 - Why is wearing a helmet a good way to manage risk? _____



5. Expected vs Possible Outcomes

For each event, write how likely it is to happen using a percentage.

Event	Likelihood (Percentage)	Explain your answer
a) Your favourite team wins today.		
b) You get to school on time.		
c) You see a rainbow tomorrow.		
d) The moon is made of cheese.		



6. Decision Making with Probability

For each decision, use probabilities to help you choose the best option.

- a) You can take a bus or walk to school.
- Bus: 70% chance of being on time
 - Walk: 90% chance of being on time
- Which option has a higher chance of you being on time? _____
- Why did you choose this option? _____



- b) You can bring one of these items on a picnic.
- Ice cream: 60% chance it will stay frozen
 - Sandwich: 95% chance it will stay good
 - Chocolate bar: 40% chance it won't melt
- Which item should you bring? Why? _____



7. Create a Risk Scale

Choose any five events from your life. Estimate the probability (as a percentage) of each happening and place them on the scale below.

Event	Probability (%) (0%–100%)	Place on the scale (write the letter)
a)		
b)		
c)		
d)		
e)		



8. Reflect and Explain

- a) Why is it important to think about the probability of events before making a decision? _____
- b) How can understanding risk help keep people safe? _____
- c) Give an example of a time when understanding probability helped you (or could help you) make a better choice. _____



I can...

- ☐ estimate and compare likelihoods.
- ☐ describe risk using probability.
- ☐ use probability to make decisions.
- ☐ explain how to manage risk.
- ☐ apply chance to real-life situations.

Chance Words



Great job!

You are using probability to make smart and safe decisions!